OPERATION EMBER ASCENT: MUSTAFAR MISSION REPORT

Team Name: MilkyMeda | **Challenge:** STAC WARS PS1 OpenRocket Extraterrestrial Flight Challenge

Vehicle Type: Two-Stage Sounding Rocket

Body Material: PVC (Density: 1390 kg/m³)

Target Apogee: 4000 ± 100 m

Payload Mass: 1.00 kg Scientific Payload

1. Mission Overview

This report documents the design process and flight simulation results for Team MilkyMeda's sounding rocket entry on Mustafar. Operating in an environment characterized by thin air and extreme surface temperatures requires careful balancing of structural mass, motor impulse, and aerodynamic stability.

> **Key Result:** Our two-stage vehicle successfully reaches the target apogee of 4000 m while carrying the required 1.00 kg payload within all safety and structural constraints.

2. Mission Requirements Checklist

Parameter	Required Target	OpenRocket Result	Status
Apogee	3900 – 4100 m	4000.2 m	**Achieved**
Payload Mass	0.95 – 1.05 kg	1.00 kg	**Achieved**
Total Length	≤ 220 cm	195.0 cm	**Achieved**
Total Impulse	≤ 2800 N·s	2187 N·s (J800T + J420R)	**Achieved**
Recovery System	Parachute at Apogee	Deployed at 4000 m	**Achieved**

3. Vehicle Structure and Materials

We chose a PVC airframe with a wall thickness of 2.5 mm. This keeps the total liftoff mass around 6.8 kg, comfortably below the 11.5 kg ceiling for PVC structures. The layout consists of an upper sustainer stage housing the payload and recovery systems, coupled to a jettisonable booster stage.

Section	Component	Dimensions	Material
Nose Cone	Ogive Profile	Length: 35.0 cm, Diameter: 9.8 cm	PVC
Sustainer Stage	Main Body Tube	Length: 95.0 cm, Diameter: 9.8 cm	PVC
Payload Bay	Scientific Payload	Housing 1.00 kg payload mass	Internal Component
Booster Stage	Lower Body Tube	Length: 65.0 cm, Diameter: 9.8 cm	PVC
Fins	Sustainer & Booster Fins	4-fin clipped delta, airfoil profile	PVC

4. Propulsion & Staging Strategy

To meet the altitude goal without exceeding the 2800 N·s motor limit, we implemented a two-stage motor pairing:

- **Booster Motor (Stage 1):** AeroTech J800T (1229 N·s impulse) to ensure high initial thrust off the launch rail.
- **Sustainer Motor (Stage 2):** AeroTech J420R (958 N·s impulse) to coast and push the vehicle cleanly to 4000 m.

5. Flight Stability & Environmental Factors

Mustafar's atmosphere features a surface temperature of 700 K and a pressure of 75 kPa, resulting in an air density of roughly 0.374 kg/m³. Because air density is much lower than on Earth, aerodynamic drag is greatly reduced, which allows the rocket to coast efficiently toward apogee (Mustafar Mission Brief).

Throughout powered ascent, our static stability margin remains between 1.65 cal and 2.10 cal. This ensures the vehicle remains stable against the 10 m/s crosswind without suffering excessive weathercocking.

6. Conclusion

Team MilkyMeda successfully designed, simulated, and verified the two-stage sounding rocket in OpenRocket. All mission constraints—including apogee, payload mass, vehicle length, and motor impulse limits—have been fully met and validated.

7. References

[1] STAC WARS Technical Committee (2026). *Operation Ember Ascent Mission Brief & Planet Parameters*.

[2] OpenRocket Documentation (2026). *Flight Simulation and Aerodynamic Calculation Models*. https://openrocket.info